

TUTORIAL – 9: Black-Box & White-Box Testing

Aim

To analyze and apply **Black-Box Testing**, **White-Box Testing**, **Integration Testing**, and **System Testing** on a sample **Student Management System (Login Module)** and document the testing process, execution, and results.

Description

Software testing ensures that an application works correctly and meets user requirements. Different testing techniques are used depending on the testing objective.

1. Black-Box Testing

Black-Box Testing is a testing technique where the tester verifies the application's functionality **without knowing the internal source code**.

The tester provides inputs and checks whether the output matches the expected result.

Characteristics

- No programming knowledge required.
- Based on Software Requirement Specification (SRS).
- Tests external behavior of the application.

Example

Input

Username = admin

Password = admin123

Expected Output

Login Successful

2. White-Box Testing

White-Box Testing is a testing technique where the tester has complete knowledge of the application's internal source code.

The tester verifies:

- Program logic
- Conditions
- Loops
- Branches
- Statements

- Code coverage

Characteristics

- Programming knowledge required.
- Checks internal logic.
- Performed mainly by developers.

3. Integration Testing

Integration Testing verifies whether multiple modules work together correctly.

Example

Modules

- Login Module
- Dashboard Module

Testing

After successful login,

Dashboard should open.

If communication between modules is successful,

Integration Test = PASS

4. System Testing

System Testing verifies the **complete application**.

The entire Student Management System is tested including:

- Login
- Registration
- Student Details
- Search
- Logout

Difference Between Black-Box and White-Box Testing

Black-Box Testing	White-Box Testing
Tests external functionality	Tests internal program logic
Source code not required	Source code required

Performed by Testers	Performed by Developers/Testers
Requirement based	Code based

Problem Statement

Perform **Black-Box Testing**, **White-Box Testing**, **Integration Testing**, and **System Testing** on the Login Module of the Student Management System. Execute test cases, analyze the results, and document the testing activities.

How to Perform This Practical (Student Procedure)

Step 1: Study the Software Requirements

Read the Login Module requirements.

Requirements

- User should login with valid credentials.
- Invalid credentials should display an error message.
- Dashboard should open after successful login.

Step 2: Perform Black-Box Testing

Without looking at the source code,

Execute the following inputs.

Username	Password	Expected Result
admin	admin123	Login Successful
admin	12345	Invalid Password
student	admin123	Invalid Username
blank	blank	Validation Message

Record

Expected Result

Actual Result

PASS / FAIL

Step 3: Perform White-Box Testing

Study the source code.

Verify

- if condition
- else condition
- branch execution
- statement execution

Check whether every line of code executes correctly.

Step 4: Perform Integration Testing

Verify communication between modules.

Flow

Login Module

↓

Dashboard Module

↓

Logout Module

Check whether data is transferred correctly.

Step 5: Perform System Testing

Test the complete application.

Modules

- Login
- Registration
- Student Details
- Search
- Logout

Verify complete application behavior.

Step 6: Record Results

Compare

Expected Result

Actual Result

Mark

PASS

or
FAIL

Step 7: Prepare Observation

Example

- ✓ Login successful.
- ✓ Dashboard opens correctly.
- ✓ No communication errors.
- ✓ All modules work properly.

Implementation Details

The Login Module of the Student Management System is tested using **Black-Box**, **White-Box**, **Integration**, and **System Testing** techniques to verify functional correctness, internal program logic, module interaction, and complete system behavior.

Parameter Details

Parameter	Details
Project Name	Student Management System
Application/Software	Student Management System
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Python
IDE/Tool Used	Visual Studio Code / PyCharm
Input/Test Data	admin/admin123, admin/12345, student/admin123

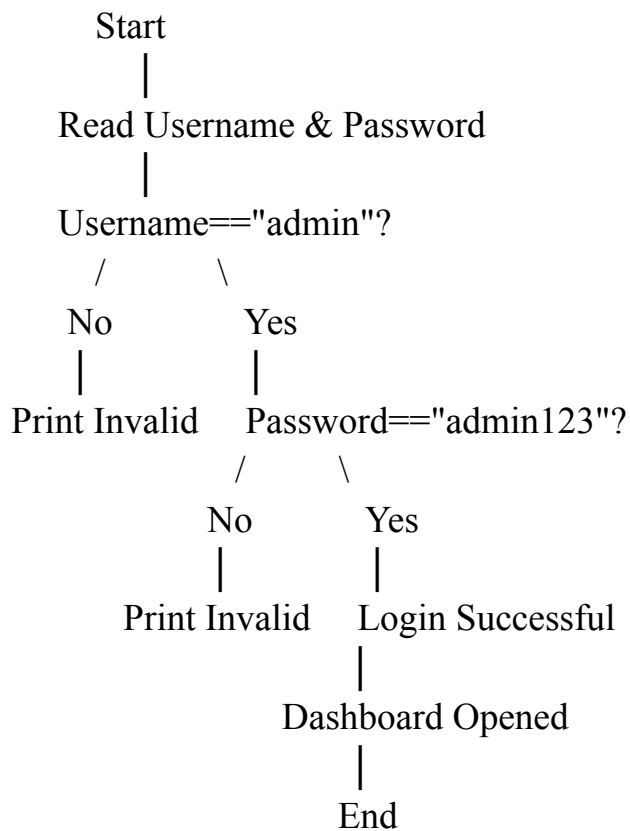
Source Code

Login Module

```
username = input("Enter Username: ")
password = input("Enter Password: ")

if username == "admin" and password == "admin123":
    print("Login Successful")
    print("Dashboard Opened")
else:
    print("Invalid Username or Password")
```

Control Flow (White-Box Testing)



Test Case Detailed

Field	Details
Test Case ID	TC001
Requirement ID	R001
Module Name	Login Module
Feature Under Test	User Authentication
Test Scenario	Login with valid username and password
Test Objective	Verify successful login
Test Type (Unit/Integration/System/UAT)	System Testing
Test Technique (Black Box/White Box)	Black Box Testing
Priority	High
Severity	Critical
Preconditions	Login page should be displayed
Postconditions	Dashboard should open
Test Environment	Windows 10
Browser/OS (if applicable)	Google Chrome
Test Data	admin / admin123
Expected Result	Login Successful and Dashboard Opens
Actual Result	Login Successful and Dashboard Opens

Status (Pass/Fail)	Pass
Executed By	Student Name
Execution Date	DD/MM/YYYY

Additional Test Cases

Test Case ID	Testing Technique	Scenario	Expected Result	Status
TC001	Black-Box	Valid Login	Login Successful	Pass
TC002	Black-Box	Invalid Password	Error Message	Pass
TC003	White-Box	Verify if-condition	Branch Executed	Pass
TC004	White-Box	Verify else-condition	Branch Executed	Pass
TC005	Integration	Login → Dashboard	Dashboard Opens	Pass
TC006	System	Complete System	All Modules Work	Pass

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Launch Application	-	Login Page	Login Page	Pass
2	Enter Username	admin	Username accepted	Username accepted	Pass

3	Enter Password	admin123	Password accepted	Password accepted	Pass
4	Click Login	Click	Dashboard Opens	Dashboard Opens	Pass
5	Logout	Click	Login Page Opens	Login Page Opens	Pass

Testing Activities Performed

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1	Requirement Analysis	Requirement Analysis	Analyze Login Requirements	SRS Document
2	Design	Test Planning	Prepare Test Plan	Test Plan
3	Development	Test Case Development	Create Black-Box & White-Box Test Cases	Test Case Document
4	Testing	Test Execution	Execute Black-Box, White-Box, Integration & System Tests	Test Execution Report
5	Testing	Test Closure	Prepare Test Summary	Test Summary Report

Observation

Sr. No.	Observation	Remarks
1	Black-Box Testing completed successfully	Pass
2	White-Box branch coverage verified	Pass

3	Login communicates correctly with Dashboard	Pass
4	Complete application tested successfully	Pass
5	No major defects found	Completed

Output

Student Management System

===== LOGIN =====

Username : admin

Password : admin123

Login Successful

Dashboard Opened

=====

Black-Box Testing : PASS

White-Box Testing : PASS

Integration Testing : PASS

System Testing : PASS

Overall Result

Testing Completed Successfully

Solve the Below Mention Questionnaires

1. Which testing technique verifies functionality without examining source code?

- A) White-Box Testing
- B) Integration Testing
- C) Black-Box Testing**
- D) Unit Testing

Answer: C) Black-Box Testing

2. Which testing technique requires knowledge of internal program code?

- A) System Testing
- B) Black-Box Testing
- C) Acceptance Testing
- D) White-Box Testing**

Answer: D) White-Box Testing

3. Integration Testing is mainly used to verify:

- A) Database Performance
- B) Communication between modules**
- C) Source Code Formatting
- D) User Interface Design

Answer: B) Communication between modules

4. System Testing validates:

- A) Individual Functions Only
- B) Single Module Performance
- C) Complete Application Behavior**
- D) Database Tables Only

Answer: C) Complete Application Behavior

5. Which testing checks execution paths and branch coverage?

- A) Black-Box Testing
- B) White-Box Testing**
- C) Regression Testing
- D) Usability Testing

Answer: B) White-Box Testing

Conclusion / Outcome

- Understood the concepts of **Black-Box Testing**, **White-Box Testing**, **Integration Testing**, and **System Testing**.
- Designed and executed test cases using both Black-Box and White-Box testing techniques.
- Verified communication between the Login and Dashboard modules through Integration Testing.
- Validated the complete Student Management System using System Testing.
- Learned how different testing techniques ensure software quality, reliability, and correctness before deployment.